Mustafa Jamshidi

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Portfolio: https://mustafajamis.github.io/DesktopLikePortfolio/

EDUCATION

Wichita State University — **Wichita, Kansas**
B.S in Computer Science — *December 2025*

EXPERIENCE

John Deere (Vehicle Diagnostic Platform Team) — **Moline, Illinois**
Full Stack Software Engineer — *Jun 2026*

- Designed and developed a TypeScript/Node.js module for CAN bus traffic recording in John Deere's vehicle diagnostics platform, replacing a legacy 5-layer C#/COM/.NET architecture with a single maintainable component.
- Diagnosed and resolved a critical process-lifecycle defect by analyzing production logs; restored reliable start/stop/restart behavior and eliminated a class of recurring recording failures.
- Established regression test coverage for the fixes and edge cases, growing the suite to 118 passing unit tests at 90%+ coverage (Vitest).
- Validated the module end-to-end against live CAN hardware using an Electron integration harness across multiple adapters and channels.
- Drove the change through peer code review and pull requests and led the setup of a CI/CD build-and-publish pipeline to an internal npm registry.

LifeSizeAgent (Whisper Voice Integration Team) — **Remote**
Team Developer — *March 2025 – Jun2025*

- Integrated Whisper AI for real-time speech-to-text, enabling users to interact with the app through voice input.
- Developed and tested text-to-speech (TTS) pipelines using Eleven Labs and prepared code for integration with the Simli avatar.
- Collaborated cross-functionally to merge key features, such as voice-to-text and avatar integration, into production branches to support smooth deployment.

NetApp (ONTAP & E- series) — **Wichita, Kansas**
Software Engineer Intern — *Dec 2022 – Jun 2024*

- Strengthened code reliability within a mature CI/CD pipeline by conducting comprehensive unit tests for C/C++ projects on RHEL systems, directly improving product safety.
- Performed system administration tasks on RHEL servers, including user/group management, setting file permissions, and debugging code to resolve system-level issues.
- Improved process uniformity and execution efficiency by 30% by leading the conversion of over 3,000 lines of legacy Perl to Python, enhancing cross-functional compatibility.
- Designed and implemented an IMT test for Dagger 2, improving system stability and watchdog functionality.
- Diagnosed and resolved critical defects in E-Series controller firmware, boosting product reliability.
- Executed stress testing to validate system stability under extreme conditions, ensuring robustness and reliability during peak loads.

Telecom Office at Wichita State University — **Wichita, Kansas**
Telecom Student Technician — *May 2022 – Dec 2022*

- Provided direct desk-side support to university staff for troubleshooting hardware (desktops, laptops, printers), software, and peripheral issues.
- Performed hands-on troubleshooting and diagnostics on network infrastructure, including executing detailed LAN port audits and testing on communication systems to ensure 100% uptime.
- Managed IT inventory and developed meticulous documentation for troubleshooting procedures, resolutions, and system workflows

PROJECTS

ReviewPilot | AI-Powered GitHub PR Reviewer
Personal Project — *Summer 2025*

- Built an AI-assisted pull request review platform that ingests GitHub webhook events, analyzes PR diffs, and generates structured code review feedback.
- Implemented rule-based security and quality checks to flag risky changes such as hard-coded secrets, SQL injection patterns, unsafe eval usage, auth-related updates, and production logging.
- Secured webhook processing with GitHub signature verification and automated posting of formatted review comments back to pull requests through the GitHub API.

USDA Rural Development Chatbot
Hackathon Project — *Fall 2025*

- Engineered a full-stack AI application with a Retrieval-Augmented Generation (RAG) pipeline using Python, Flask, and MongoDB Atlas Vector Search to provide accurate answers about USDA programs.
- Implemented a Python-based web scraper to automatically build and update the knowledge base from official USDA program websites.
- Integrated large language models (Google Gemini and a local Llama instance) to power a natural language conversational interface.

SKILLS

Programming: C/C++, JavaScript, Python, Perl, CUDA, F#, HTML/CSS, PostgreSQL, SQL, PHP, Swift, TypeScript, React Native, Go
Frameworks: React, Node.js, Flask, FastAPI, WordPress
Developer Tools: Git, Git Bash, GDB, AWS CDK, Unreal Engine 5, JIRA, Microsoft Office, VS Code, PyCharm
Operating Systems: Linux (RHEL), Windows, MAC